NegLaB Research Colloquium – October 13, 2026

# Conjunction, negation, and how focus affects their scope

## When De Morgan is not the answer

Andrea Matticchio

UMass Amherst & Goethe University Frankfurt

# 1 | Relevance

2 Negated conjunction

3 Focus and alternatives

4 Modal intervention

5 Conclusions

# Question-answer congruence

A puzzle for semantics: Which sentences answer which questions?

- Classical answer: **relevance**; *Lewis 1988; Groenendijk & Stokhof 1984; Roberts 2012*
- Still the main factor in recent work. *Feinmann 2026; Benbaji-Elhadad & Doron 2026*

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**Today's thesis**

Relevance is not what grammar uses to decide congruence.

# Questions partition the logical space

Two assumptions:

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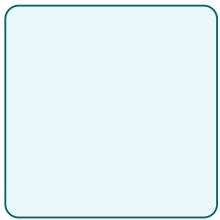
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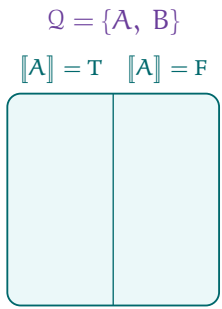
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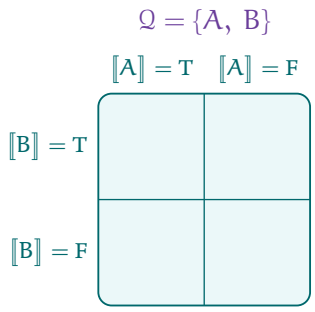
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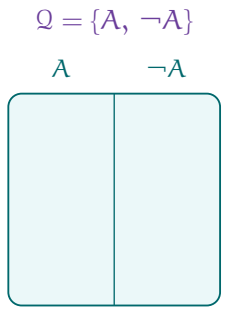
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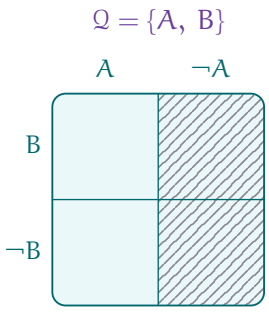
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**One of many definitions**

P is relevant to Q iff P rules out at least one whole cell of Q. *Groenendijk & Stokhof 1984*



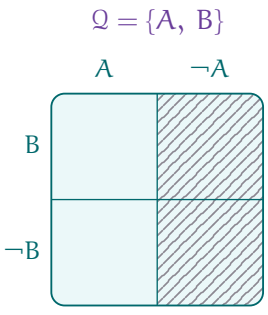

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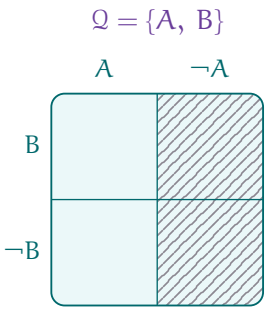

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   *a or b* answers *a, b, or c*?



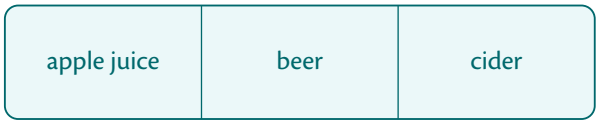

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# No need to pick just one cell

CONTEXT: *Apple juice, beer, cider. Taro had exactly one drink.*

(1) Q: Which drink did Taro have?

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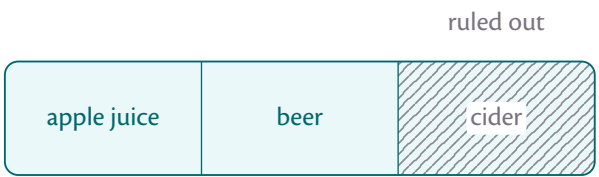


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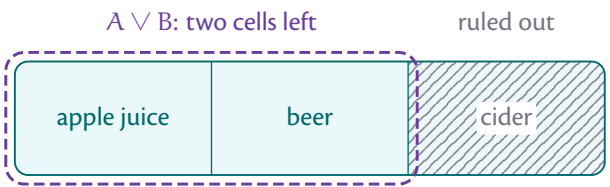
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relevant without resolving the question

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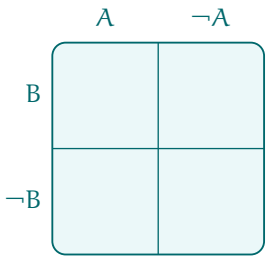
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# De Morgan

A classical equivalence: negated conjunction has the same truth conditions as a disjunction.

(2) Taro didn't have both the apple juice and the beer. $\neg(A \wedge B)$

(3) Either Taro didn't have the apple juice, or he didn't have the beer. $\neg A \vee \neg B$




using *both* to remove the homogeneous reading of conjunction

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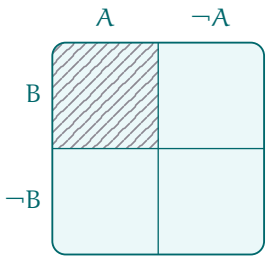
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$$\neg(A \wedge B) \iff \neg A \vee \neg B$$

using *both* to remove the homogeneous reading of conjunction




same cell ruled out

# Different answers despite being equivalent

(4) Q: Which drinks did Taro not have?

A: Either he didn't have [the apple juice]$_F$, or he didn't have [the beer]$_F$. $\neg A \vee \neg B$

A′: He didn't have both [the apple juice]$_F$ and [the beer]$_F$.

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✓ $\neg A \wedge \neg B$

**Intuition**

A′ only has a reading where conjunction scopes above negation.

# The experiment

- Acceptability judgment task in Italian, 1–7 Likert scale;
- $2 \times 2$ design, within participants and within items;
- 38 participants, 36 items.

*joint work with Maik Thalmann (Ruhr University Bochum)*

| | polarQ | whQ |
|---|---|---|
| nand | Did he do both A and B, or not? | Which ones did he not do? |
| or | Did he do A or B, or not? | Which ones did he do? |

with ignorance continuations, which remove the $\neg A \wedge \neg B$ reading

# Items

(5) a. Did Peter hide both the necklace and the bracelet, or did he not? polarQ-nand
No, he didn't hide both the necklace and the bracelet, but I don't remember which he didn't hide.

b. Which pieces of jewelry did Peter not hide? whQ-nand
He didn't hide both the necklace and the bracelet, but I don't remember which he didn't hide.

c. Did Peter hide the necklace or the bracelet, or did he not? polarQ-or
Yes, he did hide the necklace or the bracelet, but I don't remember which he hid.

d. Which pieces of jewelry did Peter hide? whQ-or
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# An example trial

Quanto è accettabile la risposta alla domanda?

*C'erano tre gioielli sul comodino: una collana, un braccialetto e un anello.*
*Pietro, che aveva intenzione di nascondere alcuni dei gioielli, ma non di nasconderli tutti, riceverà una ricompensa se ha nascosto sia la collana che il braccialetto.*



Ma quindi Pietro ha nascosto sia la collana che il braccialetto, oppure no?

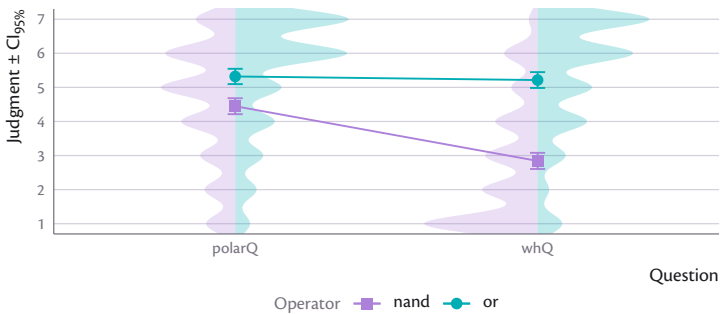
**No, non ha nascosto sia la collana che il braccialetto, ma non ricordo quale non abbia nascosto.**



(per nulla accettabile) 1 2 3 4 5 6 7 (totalmente accettabile)

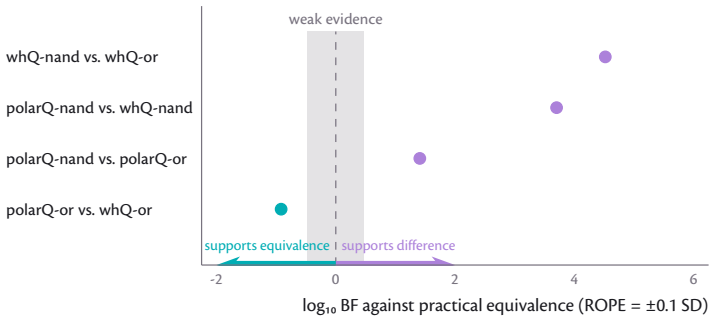
polarQ-nand condition

# Results



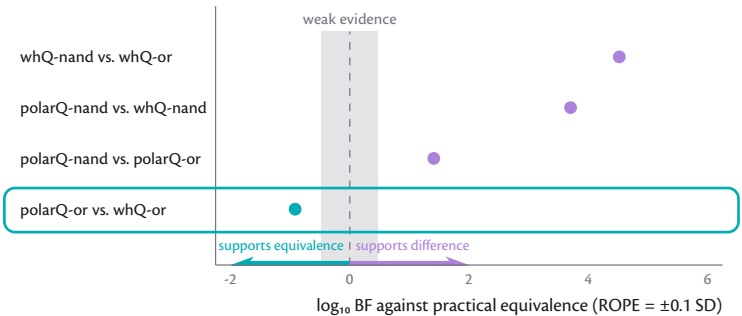

raw results; shaded areas are rating distributions

# Bayes factors




cumulative probit model, maximal random effects for participants and items

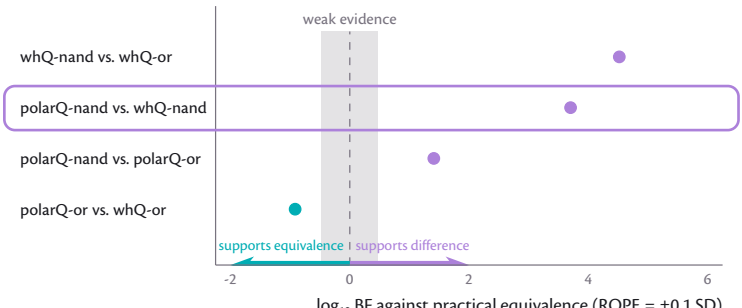
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*or*:
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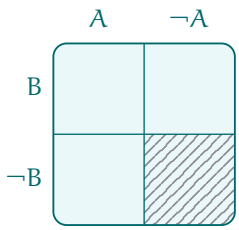
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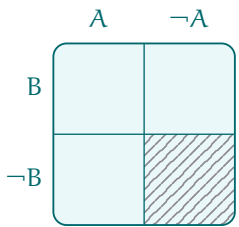
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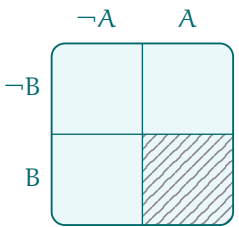

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answer: $A \vee B$

- *Which ones did he hide?* $A \vee B$ is relevant: it rules out one cell;

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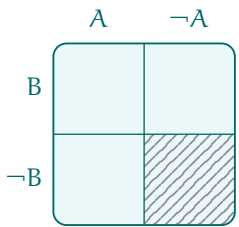

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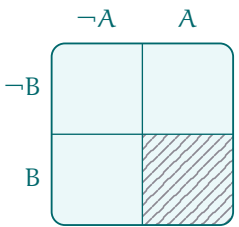

$\mathcal{Q} = \{\neg A, \neg B\}$
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- *Which ones did he hide?* $A \vee B$ is relevant: it rules out one cell;
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- *Which ones did he hide?* $A \vee B$ is relevant: it rules out one cell;
- *Which ones did he not hide?* $\neg(A \wedge B)$ is relevant in the same way;
- But negated conjunction is unable to answer a *wh*-question the way disjunction can.

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# Two focus placements for negated conjunction

(6) He didn't have both [the apple juice]$_F$ and [the beer]$_F$.

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- Focus on the conjuncts: only $\neg A \wedge \neg B$;
- Polarity focus: $\neg(A \wedge B)$ is fine.

The experiment shows that negation can scope over conjunction with polar questions.

# Focus: a second dimension of meaning

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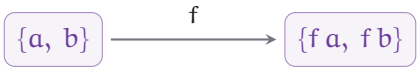
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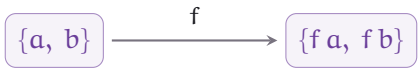


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- The alternatives at the root are the question the sentence answers.



# Polarity focus

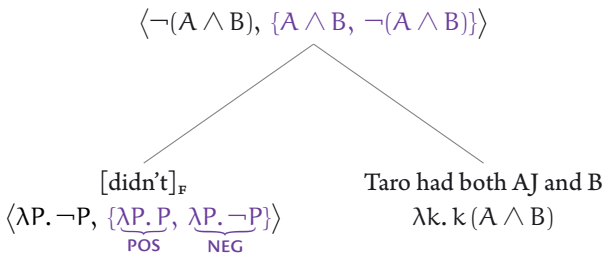
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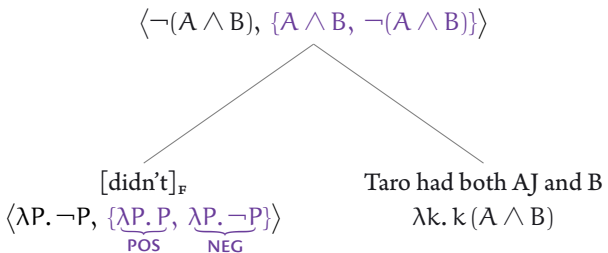
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Taro had both AJ and B
$$\lambda k.\, k\,(A \wedge B)$$

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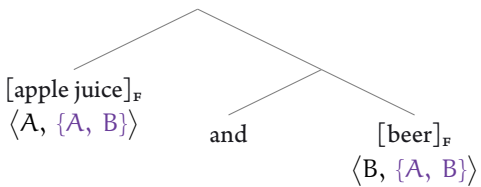


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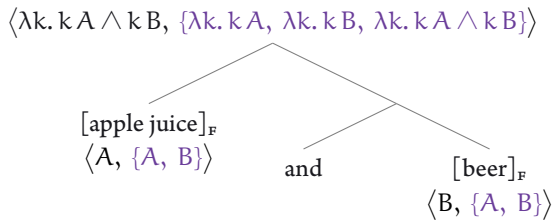



alternatives $\{A \wedge B, \neg(A \wedge B)\}$: a polar question ✓

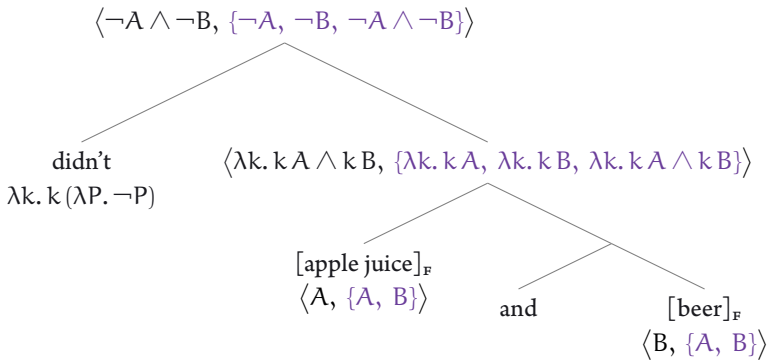
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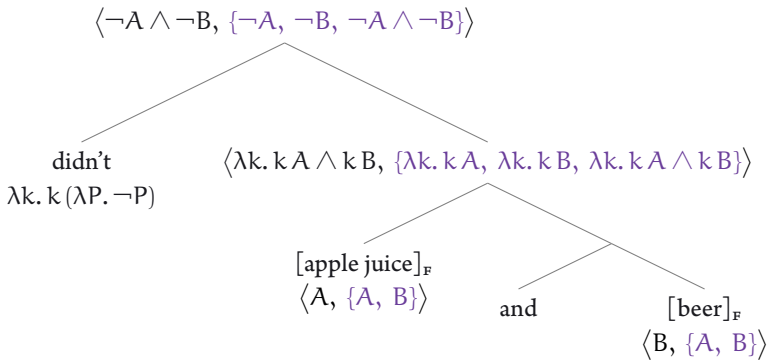
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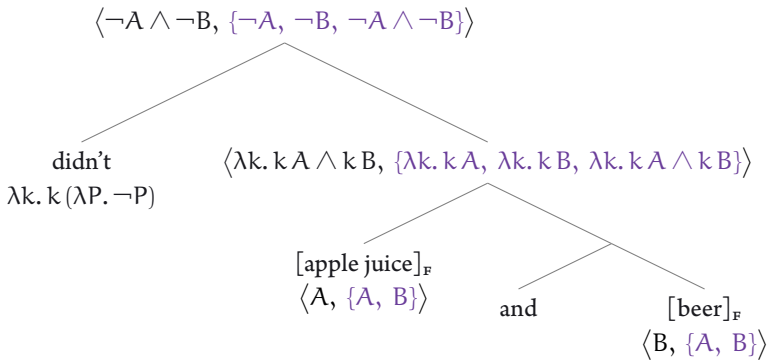
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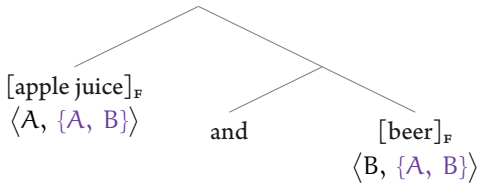
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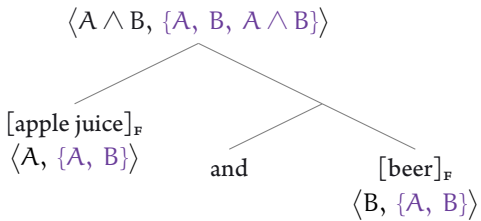

wide scope of conjunction

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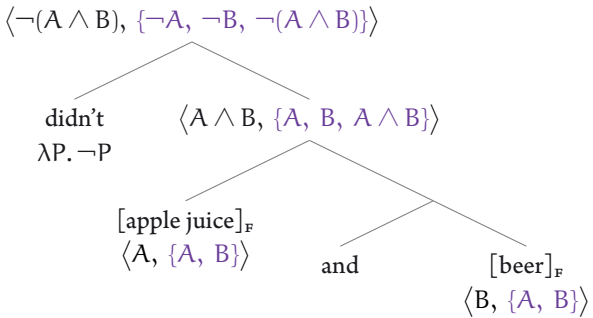
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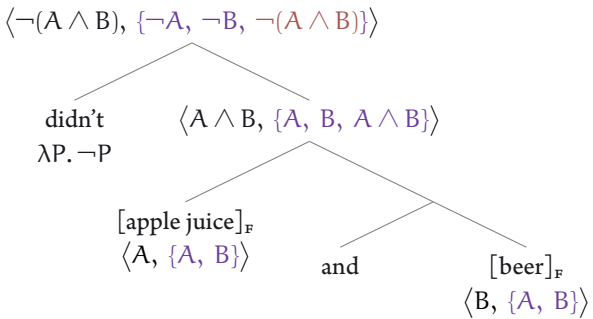
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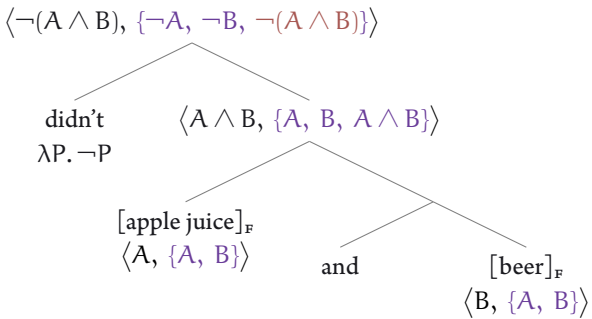


# Focus on the conjuncts: narrow scope




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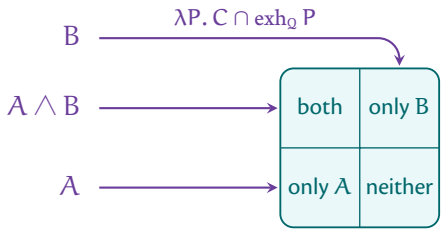
Is this still an answer to the *wh*-question?

# Question-partition matching

**QPM:** A question is well-formed only if exhaustifying each alternative in the context C picks out a cell of its own, and every cell is picked out this way. *Fox 2018*

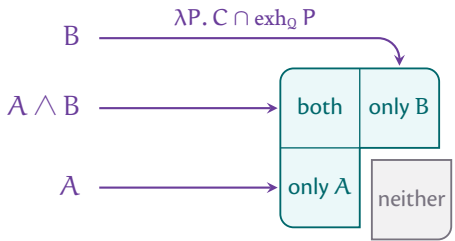
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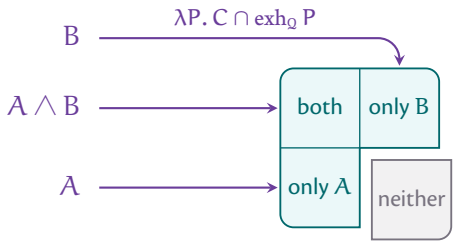



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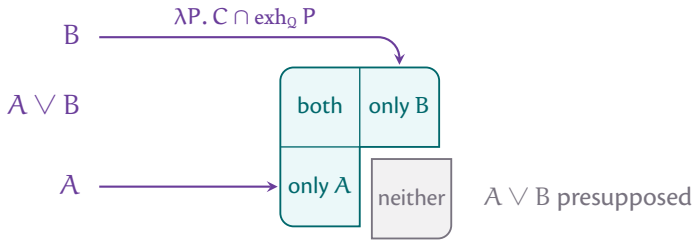

*Did Taro have the apple juice, the beer, or both?*

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Focus alternatives have to match a question, so QPM constrains them too.

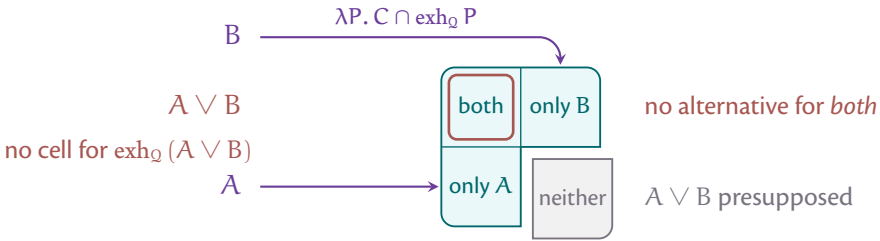
# A case where matching fails

Adding disjunctions of alternatives to the set makes QPM fail.



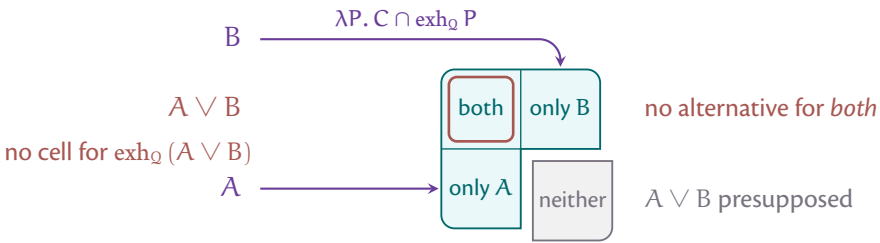
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Adding disjunctions of alternatives to the set makes QPM fail.




This can make negated conjunction fail QPM:

$$\{\neg A, \neg B, \neg(A \wedge B)\} = \{\neg A, \neg B, \neg A \vee \neg B\}$$

# Four derivations

| | scope and focus | alternatives | QPM |
|---|---|---|---|
| (a) | $[AJ]_F$ and $[B]_F$ | $\{A,\ B,\ A \wedge B\}$ | ✓ |

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- $\neg(A \wedge B)$ only via (d): an answer to the polar question;
- (c) does not match any question that passes QPM.

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*or* $\left\{\right.$ $\langle A, \{A,\ B,\ A \wedge B\}\rangle$ ✓ QPM

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Each is a good answer to the same question; $A \vee B$ is never an alternative.

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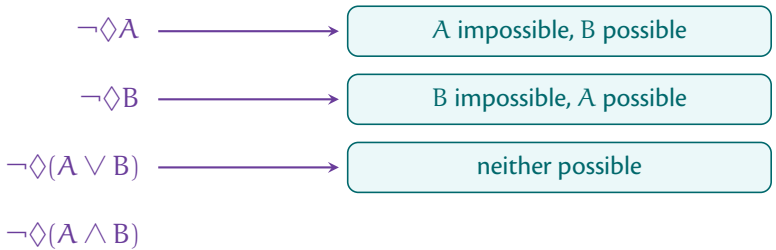
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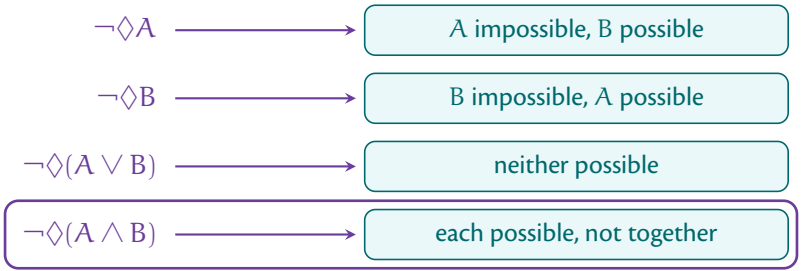
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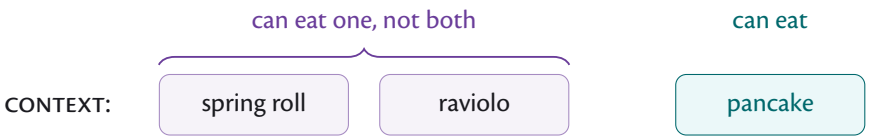


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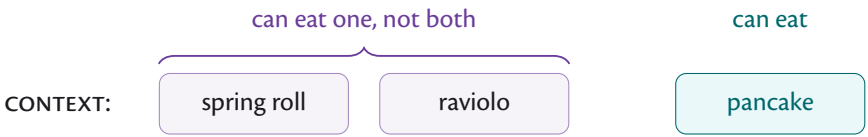
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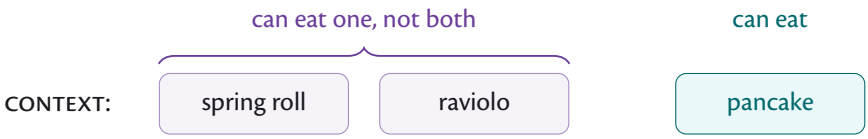


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(8) Q: Which leftovers is Peter not allowed to eat?
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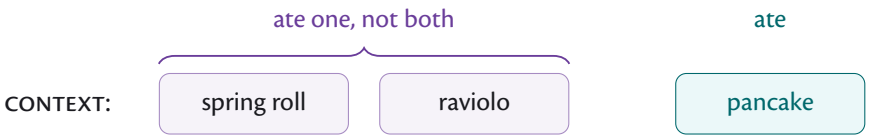



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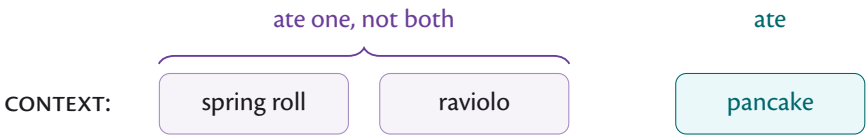
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predicted ✓

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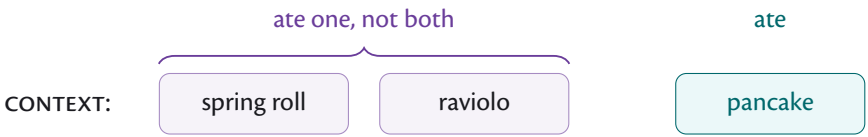
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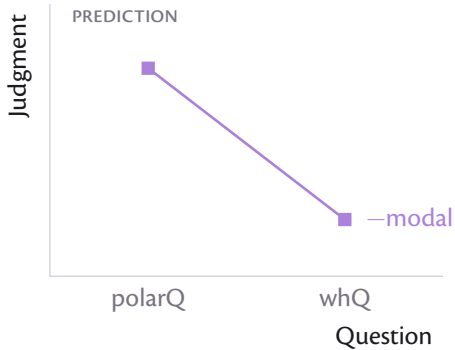

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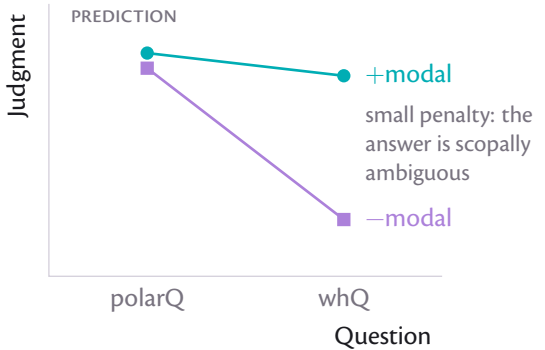
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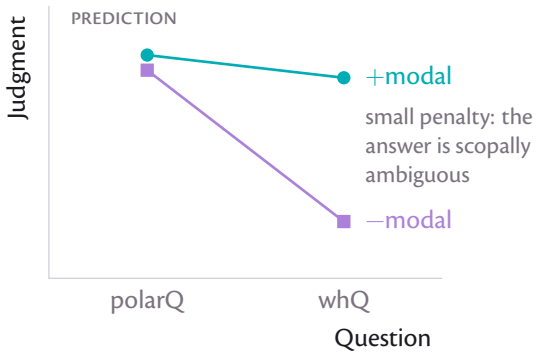
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The predicted contrast is hard to derive without structured answers.

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to be tested soon in project B04

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# Model coefficients

